

Getting Started with Protractor



Protractor is an end-to-end test framework for Angular and AngularJS applications. It can run tests for your applications in a real browser, by interacting with it as a real user.

Before getting started with Protractor, you must know why you are going to use it. Angular applications have a lot running in the background without any visible behaviour change in the browser, which makes it difficult for Selenium to understand and synchronise them. Thus, scripts written in Selenium have clumsy waits and sleeps. Protractor was made by the people who created Angular. It was developed to capture this hidden background behaviour and to test Angular applications.

Protractor

Protractor is an end-to-end test framework for AngularJS applications. It is a wrapper around WebDriverJS and supports behaviour-driven development frameworks. It acts as an abstraction layer to automate *write* and *execute* easily for Angular and AngularJS applications. It is used to test whether the flow of an AngularJS application is as designed from start to end, based on the system dependencies or environment. It is intended not only to test AngularJS applications but also to write automated regression tests for normal Web applications.

When and where to use it

AngularJS applications have some extra HTML attributes like *ng-repeater*, *ng-controller*, *ng-model*, etc, which are not included in Selenium locators. Selenium is not able to identify these Web elements using its own code. Protractor on the top of Selenium can handle and control these attributes in Web applications. Basically, Protractor was developed for only Angular applications but has now been enhanced to support non-Angular applications also.

Process flow

The workflow of Protractor is quite easy. It just needs two files *conf.js* and *spec.js*. The first file provides all that's related to the configuration, while *spec.js* is a file that specifies how and where the test should run.

The high-level workflow in Figure 1 shows how the combination of *conf.js* and *spec.js* is used by the Protractor runner class to execute the test. It is then run by the Selenium server in the browser, as specified in the *spec.js* file.

Figure 2 depicts the interaction of Protractor with AngularJS.